Unit V: Plant Physiology (Functional Organisation)

Chapter 11

Transport in Plants

Learning Objectives

The learner will be able to,

- *Recall knowledge of basic physical and biological processes studied in previous classes.*
- *Classify, differentiate and compare the process of active and passive transport.*
- *Understand the mechanism of absorption of water.*
- *Analyse the various theories in ascent of sap.*
- *Understand the process of transpiration and Compare the various types of transpiration.*
- *Discuss the mechanism of phloem translocation.*
- *Understand the process behind mineral absorption.*

Chapter Outline

11.1 Types of transport
11.2 Cell to Cell transport
11.3 Plant water relations
11.4 Absorption of water
11.5 Ascent of Sap
11.6 Transpiration
11.7 Translocation of organic solutes
11.8 Mineral absorption

Over 450 million years ago (the Ordovician period in Paleozoic era) plants migrated from their own sophisticated water world to newly formed land. The land had harsh environment; water availability was deeper and so plants struggled for getting water for their very existence. Some of them failed to survive and rest adopted themselves to the new world. The biggest adaptations followed for their survival was building their own water absorbing systems to draw water from deep inside the land. The creation and updating of water absorbing system (vascular tissues) led to the diversity of the plant kingdom. The gregarious growth of prehistoric pteridophytes, gymnosperms and present-day flowering plants led to the biggest challenge in the transport of water from root to several meters high trees against gravity. In this chapter, we will study the events taking place between the gain of water in roots and loss in leaves and the mechanisms behind the basic physical and biological processes in the movement of water, gases and minerals in plants. Further, we study how food material synthesized in the leaf can be transported to various utilizing and storage areas against struggles and challenges.

11.1 Types of Transport

Transport is the process of moving water, minerals and food to all parts of the plant body. Conducting tissues such as xylem and phloem play an important role in this.

What is the need for transport? Water absorbed from roots must travel up to leaves by xylem for food preparation by photosynthesis. Likewise, food prepared from leaves has to travel to all parts of the plant including roots. Both the processes are interconnected and depend on each other.

❖ Based on the distance travelled by water (sap) or food (solute) they are classified as (a) Short distance (Cell to cell transport) and (b) Long distance transport.

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